

Supervised and Self-Supervised Learning on Food-251

FoodCNN under a 10M-parameter budget on a single 8GB GPU

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The problem

Supervised and Self-Supervised Learning on Food-251

FoodX-251, under two hard constraints

- 251 fine-grained food categories, web-crawled **noisy** training labels, small clean validation set (Kaur et al., 2019)
- Fine-grained: inter-class differences (plating, garnish) can be **subtler** than intra-class variation (lighting, angle, portioning)
- Two constraints, not chosen for effect:
 - **From scratch** – no ImageNet or other pretraining
 - **Under 10,000,000 trainable parameters**, on a single **8GB** laptop GPU
- These two constraints shape every decision that follows

What's actually long-tailed here? Not the classes.

- Median 471 images/class, p10/p90 366/580 – close to **balanced**
- The famous 19.3x imbalance ratio is **one** 34-image outlier class
- **Class-balanced losses, logit adjustment, long-tail training: out of scope** – the real distributional problem is **label noise** (web-crawled train, clean val), not long tails

Protocol

Splitting the validation set, once, deterministically

- 11,994 val images -> 50/50 stratified split, fixed seed 251, every class within 1 image of exact half
- val-dev (6,063): every ablation, every architecture comparison
- val-test (5,931): touched **exactly once**, for the headline number
- Without this split, the headline number is selected-on
- 15-epoch fixed-seed proxy on val-dev (~18 min) stands in for a full 90-epoch run (~2h) during search; winner gets one full confirmation run

Architecture

FoodCNN: spend the budget where it moves accuracy

- Stem (3x3 s2 conv + BN + ReLU + 3x3 s2 maxpool) -> 1/4 resolution
- 4 residual stages, doubling width each stage: **64-128-256-512**, block counts [2, 2, 2, 1]
- BN after **every** conv incl. shortcut projections; zero-init the 2nd BN's γ – each block starts as identity, safe for a high LR
- ReLU **after** the residual add

The one fix that mattered most:

- Pool to (1, 1), not (7, 7)
- (7, 7) map -> flatten -> Linear costs **6.4M params** (94% of the whole model) on the **least** expressive layer
- Global pool first -> Linear(512, 251) = **129k params**
- Result: **6.58M params**, 65.8% of budget, 98% inside the trunk

Search by throughput and memory first, accuracy second

Screened trunk variants **synthetically** (no GPU-hours on accuracy yet):

- Stage widths must be **multiples of 32** – an off-alignment trunk has **fewer** params yet runs **46% slower** (falls off the tensor-core fast path)
- Final spatial resolution dominates cost more than any single knob – keeping a 6x6 map instead of 3x3 at **equal params** costs **2.9x** throughput and **2x** memory
- Lesson: hardware-aware screening before spending a single accuracy run

Four candidates, one seed each

Variant	Params	top-1	top-5	Peak VRAM
[2,2,2,2] 64-448	9.46M	49.99%	78.29%	2.04 GiB
[2,2,4,1] 64-512	8.94M	49.73%	77.93%	2.14 GiB
baseline [2,2,2,1] 64-512	6.58M	48.82%	77.12%	1.99 GiB
5-stage 48-512	9.08M	47.42%	75.94%	1.61 GiB

- Top three within **1.2 points** – not enough to call on one seed
- Adopt **baseline**: statistically indistinguishable, **30% fewer params**, that headroom hadn't earned its keep yet
- A follow-up sweep **below** baseline found a dead heat down to 5.18M ($p=1.0000$) and a cliff before 1.68M – capacity was never the binding constraint near baseline

Training recipe

Recipe: six axes swept, plain wins

Recipe, free at this budget:

- bf16 autocast, no GradScaler
- channels_last, cosine LR + 5-epoch warmup
- label smoothing 0.1, SGD+Nesterov
- 176px crop, 8 workers, no DataParallel
- LR search: 0.05 -> **0.8** (diminishing returns each doubling)

Ablated, one seed each, plain recipe wins:

- TrivialAugment / RandAugment: **lose** at matched epochs (RandAugment's 0.84-pt gap: **not** significant, $p=0.075$)
- Mixup, CutMix, EMA, GCE loss: **lose decisively** ($p < 0.0001$ each)
- Batch size 160-512: no lever, throughput **and** accuracy flat
- RandomResizedCrop scale floor: closes train/val gap but **loses** accuracy – it's doing double duty as regularizer

Recipe: six axes swept, plain wins

Carries forward: plain label-smoothed CE, lr=0.8, batch 256.

Results

The full run

90 epochs, baseline trunk, tuned recipe, FixRes-style eval (train 176px, test 224px center crop):

Split	top-1	top-3	top-5
val-dev, best	64.36%	82.22%	87.51%
val-test (headline, touched once)	63.83%	81.79%	87.39%

- Train/val gap closes from **+11 points** at epoch 15 to **near zero** at epoch 90 – the model was underfitting, not overfitting; it needed the full schedule, not more regularization
- 2h05m wall clock, 1.99 GiB peak VRAM – comfortably under the 8GB ceiling

Self-supervised track -- in progress

- **Plan (decided):** SimSiam pretraining, 200 epochs @ 176px (~8 GPU-h), finetune (~1.8 GPU-h), against an **equal-GPU-hour supervised control** (~8 GPU-h) – without the control, “SSL helped” is indistinguishable from “more epochs helped, regardless of objective”
- SimSiam over BYOL: no momentum encoder, no second ~6.5M-param shadow network to fit in 8GB
- SimCLR excluded: degrades below batch sizes this budget can reach
- Scaffolding lands: `src/simsiam.py`, `--init-encoder`, tested on synthetic tensors – **the run itself has not happened yet**

Discussion

Hardware measurement mattered as much as architecture

- Channel widths, spatial resolution, and **thermal throttling** moved results as much as any architectural choice
- GPU runs against a ~50W power cap; a short benchmark swung 45% between sweeps before fixing the timing methodology
- The full 90-epoch run itself ran **28% slower** than its own synthetic estimate – sustained load heat-soaks a laptop GPU in a way no short benchmark window reproduces
- Every throughput number here is an **upper bound**, and a **ratio** between variants – not a wall-clock promise

Conclusion

- A plain residual CNN with a genuinely global pooling head reaches **63.83% top-1 / 87.39% top-5** on FoodX-251, from scratch, under 10M params, on a single 8GB GPU
- Two attractive-on-paper ideas – a fifth residual stage, a depthwise-separable trunk – both **measurably lost** once actually run
- Treating wall-clock and memory as design constraints, not just the parameter cap, is what got here
- **Open:** the self-supervised comparison against an equal-compute supervised control

Questions?